

Customer Churn and Retention Analysis

IBM Telco Customer Churn Dataset | R | Customer Analytics

Objective

Identify customer characteristics associated with churn and translate the findings into practical retention insights.

Dataset

IBM Telco Customer Churn dataset, containing 7,043 customer records. The analysis focuses on contract type, payment method, tenure and monthly charges.

Analysis

The R workflow imports the customer data, prepares categorical variables, calculates churn rates by contract and payment method, compares tenure and monthly charges by churn status, and uses chi-square tests for categorical associations.

Business interpretation

Contract and payment-method differences describe customer commitment and billing behaviour. Tenure provides a customer-lifecycle perspective, while monthly charges indicate the commercial value of customers who leave.

Recommendations

Prioritise retention activity toward segments with higher observed churn. Possible actions include onboarding support, contract incentives, payment-method education and proactive service outreach. These associations should not be interpreted as causal without further testing.

Reproducibility

The repository contains analysis.R and supporting visualisations. The original dataset is expected locally and is not redistributed by the project.